

The background of the slide is a light gray gradient. It is decorated with numerous 3D-rendered organic shapes and spheres. These shapes are primarily in shades of blue and purple, with some having a gradient from light blue to white. The shapes vary in size and complexity, some resembling bubbles or fluid droplets. They are scattered across the slide, with a higher concentration in the center and right side. The text "Monitoring Redshift" is centered over these shapes.

Monitoring Redshift

Agenda

- Cluster Performance
- Query Monitoring
- System Views
- Configuration Parameters

Monitoring Redshift

- Cluster Performance (Hardware)
 - Using CloudWatch Metrics
 - Using Redshift console
- Query Monitoring
 - Using system views
 - Using Redshift console

System Views and Tables

- STV – Shows the snapshot of current system data. Stored in-memory, not persisted on disk.
- STL – Views. Provides a history of the system. Created from system logs generated by Redshift.
- SYS – Views. Used to monitor query and workload usage.
- SVV – Views. Information about database objects.
- SVCS – Views. Provides information about queries on main cluster and concurrency scaling cluster.
- SVL – Views. Provides details about queries on main cluster.

STV Views

- STV_BLOCKLIST – Contains the number of 1 MB disk blocks that are used by each slice, table, or column in a database.
- STV_LOAD_STATE – state of ongoing COPY statements.
- STV_LOCKS – Current locks in the system.
- STV_MV_INFO – Materialized view info, whether the data is stale, and state information.
- STV_SESSIONS – Active user sessions.
- STV_SLICES – Mapping of slice to node.
- STV_WLM_QMR_CONFIG – Configuration for WLM query monitoring rules
- STV_WLM_QUERY_QUEUE_STATE – Current state of the query queues for the service classes.
- STV_WLM_QUERY_STATE – Current state of queries.

STL Views

- STL_ANALYZE – Contains details of ANALYZE operations.
- STL_CONNECTION_LOG – Connection log information (Audit Logging).
- STL_EXPLAIN – EXPLAIN plan of query that has been submitted for execution.
- **STL_QUERY** – Execution information about an SQL query.
- **STL_QUERY_METRICS** – Contains query details like no. of rows processed, CPU usage, IO, disk use.
- STL_BCAST, STL_DELETE, STL_MERGE, STL_SCAN etc.

SYS Views

- SYS_ANALYZE_HISTORY – Logs details for ANALYZE operations.
- SYS_AUTO_TABLE_OPTIMIZATION - Records automated actions taken by Redshift on tables defined for automatic optimization.
- SYS_CONNECTION_LOG - Logs authentication attempts and connections and disconnections.
- SYS_LOAD_DETAIL – Records progress of each data file as it is loaded into a table.
- SYS_LOAD_HISTORY – Details of COPY commands.
- SYS_MV_REFRESH_HISTORY – Refresh history of all materialized views.
- SYS_MV_STATE - State of all materialized views. Includes base table information, schema properties, and information about recent events, like dropping a column.
- SYS_QUERY_DETAIL – Details for queries at a step level.

SYS Views

- SYS_QUERY_HISTORY – Details of user queries.
- SYS_QUERY_TEXT – Query text of all queries.
- SYS_SERVERLESS_USAGE – Details of Redshift Serverless usage of resources.
- SYS_STREAM_SCAN_STATES – Records scan states for records loaded via streaming ingestion.
- SYS_TRANSACTION_HISTORY - Details of a transaction when tracking a query.
- SYS_UNLOAD_DETAIL – View details of an UNLOAD operation. SYS_UNLOAD_HISTORY.
- SYS_USERLOG
- SYS_VACUUM_HISTORY to view details of vacuum queries.

SVV Views

- SVV_ALL_COLUMNS, SVV_ALL_SCHEMAS, SVV_ALL_TABLES.
- SVV_COLUMN_PRIVILEGES – View column permissions that are granted to users, roles, and groups.
- SVV_DATABASE_PRIVILEGES – View the database permissions that are granted to users, roles, and groups.
- SVV_DATASHARE_PRIVILEGES – Datashare permissions that are granted to users, roles, and groups.
- SVV_DATASHARES – List of datashares created on the cluster, and datashares shared with the cluster.
- SVV_DATASHARE_CONSUMERS – List of consumers for a datashare created on a cluster.
- SVV_DATASHARE_OBJECTS to – List of objects in all datashares created on the cluster or shared with the cluster.

SVV Views

- SVV_DISKUSAGE - contains information about data allocation for the tables in a database.
- SVV_EXTERNAL_COLUMNS/TABLES/DATABASES/SCHEMAS.
- SVV_MASKING_POLICY to view all masking policies created on the cluster.
- SVV_MV_INFO table contains a row for every materialized view, whether the data is stale, and state information.
- SVV_MV_DEPENDENCY table shows the dependencies of materialized views on other materialized views within Amazon Redshift.
- SVV_QUERY_STATE to view information about the runtime of currently running queries.
- SVV_RLS_POLICY – list of all row-level security policies.
- SVV_RLS_RELATION to view a list of all relations that are RLS-protected.
- SVV_ROLES

SVV Views

- SVV_ROLE_GRANTS – List of roles that are granted roles in the cluster.
- SVV_SCHEMA_PRIVILEGES – Check schema permissions that are granted to users, roles, and groups.
- SVV_TABLE_INFO - Check compression encoding, distribution keys, sort style, data distribution skew, table size and statistics etc. for tables.
- SVV_TABLES – View tables in local and external catalogs.
- SVV_TRANSACTIONS – Check open transactions and lock contention issues.
- SVV_USER_INFO - Created users.
- SVV_VACUUM_SUMMARY – Summary information about vacuum operations logged by the system.
- SVV_VACUUM_PROGRESS – Returns an estimate of how much time it will take to complete a vacuum operation that is currently in progress.

SVCS Views

- SVCS_CONCURRENCY_SCALING_USAGE – Records the usage periods for concurrency scaling. Each usage period is a consecutive duration where an individual concurrency scaling cluster is actively processing queries.
- SVCS_QUERY_SUMMARY – Execution of queries aggregated from all nodes.

SVL Views

- SVL_FEDERATED_QUERY – View information about federated query calls.
- SVL_MV_REFRESH_STATUS – Contains a row for the refresh activity of materialized views.
- SVL_QLOG – Contains a log of all queries run against the database.
- SVL_VACUUM_PERCENTAGE – Reports percentage of data blocks allocated to a table after performing a vacuum.

Configuration Parameters

- SET <configuration_parameter> TO <value>
- analyze_threshold_percent – 0 to 100 (default is 10). Redshift skips ANALYZE for any table that has a lower percentage of changed rows than specified by this parameter.
- datestyle – Can be used to set the date format. Default is ISO YYYY-MM-DD HH:MM:SS.
- enable_result_cache_for_session – Enable or disable result caching.
- max_concurrency_scaling_clusters – Set maximum concurrency scaling clusters at session level.
- query_group – Set query group for queries. Used in WLM query queues.
- search_path
- wlm_query_slot_count – Set the no. of query slots (concurrency level) for a query.
 - If the value of wlm_query_slot_count is larger than the no. of concurrency level for the query queue, the query fails.

The background is a light gray gradient. On the right side, there is a large, complex, organic shape in shades of blue and purple, resembling a splash or a cluster of bubbles. Several smaller, smooth spheres in similar colors are scattered across the background, some on the left and some on the right.

Thank you

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